Kennedy Michael (Orcid ID: 0000-0001-5676-2010)

Accepted Article

TITLE: CANNABIS, CANNABIDIOL AND TETRAHYDROCANNABINOL IN SPORT: AN OVERVIEW

AUTHOR: Michael Kennedy

AFFILIATIONS: University of New South Wales (Sydney), Department of Medicine and Clinical Pharmacology & Toxicology, St Vincent's Hospital Darlinghurst.

CONTACT AUTHOR EMAIL: drmkenn@ozemail.com.au

FUNDING: None

COI: None

ACKNOWLEDGEMENTS: Library services, UNSW

CANNABIS, CANNABIDIOL AND TETRAHYDROCANNABINOL IN SPORT: AN OVERVIEW

ABSTRACT

The World Antidoping Agency now allows cannabidiol (CBD) to be taken in elite sport but has retained all other cannabinoids on its banned list. This, together with liberalisation of cannabis laws have led to an increasing interest in its use in sport. This includes commercial sponsorship of sporting events and highlights the need for accurate information to be readily available to athletes and their advisers. This narrative review confirms that CBD and tetrahydrocannabinol (THC) do not enhance performance. CBD possesses anti-inflammatory and analgesic properties which have not been adequately evaluated in the area of sport. Some formulations of CBD contain THC or other cannabinoids that may result in a positive drug test.

Relaxation of laws in the United States and elsewhere relating to cannabis have heightened interest in its potential for use in sport and exercise. In 2016 Australia the Federal Government legalised access to medicinal cannabis and in December 2020 amended legislation to allow formulations of 98% pure botanical cannabidiol to be placed in Schedule 3 thereby allowing direct consumer purchase from a pharmacy.[1,2]

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/imj.15724

This article is protected by copyright. All rights reserved.

Cannabis was one of the many components of the Olympic Victors Dark ointment used in the Ancient Olympics for the relief of pain and swellings.[3] The present use of cannabis in sport is largely determined by peer contact, information on the internet and other promotional activity.[4,5.] Numerous high-profile sporting identities in the USA promote cannabis as an adjunct to training, to improve sleep patterns, prevent neurotrauma or relieve musculoskeletal and other discomforts. These include Ross Rebagliti Olympic snow-boarding gold medalist, USA Triatholon partnering with Pure Spectrum CBD and World Surf League appointing cbdMD as official sponsor of the 2019/2020 Jaws Big Wave Championships.[6,7 8] Efforts to evaluate its effect on performance and its usefulness as a companion to sport are complicated by variations in doses, formulations, multiple means of drug delivery, user sophistication and multiple active cannabinoids present in the preparations and their complex pharmacology. Literature searches for this narrative review included pub med, medline, Google and a large collection of reprints and proceedings of conferences held by the author.

Pharmacology of cannabinoids

In addition to the naturally occurring *cannabis sativa*, *indica* and *ruderalis* there are hundreds of cultivated hybrids and large numbers of synthetic analog molecules. *Cannabis sativa* contains over 66 cannabinoids of which nine belong to the Δ9 group. Cannabis also contains flavinoids and pyrolysis of cannabis produces about 400 products of combustion. Of the cannabinoid acids Δ9 tetrahydrocannabinol (THC), cannabidiol (CBD), cannabichrome and cannabigerol (CBG) are quantitatively the most important. While there is some variation depending on the age of the plant, botanical THC is largely in the inactive form of THC-COOH with pyrolysis releasing the active molecule.[9] THC is a partial agonist at central CB1 and peripheral CB2 receptors and is the principal agent causing psychological effects. When inhaled by an experienced user about 10-35% is bioavailable and when taken orally it has a low formulation dependent bioavailability of around 10%. Intravenously THC is largely metabolised by CYP 2C9 and CYP3A4 to predominantly 11—OH THC which has similar psychotropic properties but is present in much lower concentrations. THC concentrations fall rapidly but the terminal $t_{½}$ ß is up to 57 hours in non-users and shorter half-lives in chronic users.[10] Cannabidiol is non-psychotropic, has an inhalational bioavailability of 31% and terminal $t_{½}$ of 24-31 hours. It is largely metabolised by CYP 2C19 and CYP3A4. Oral bioavailability is also low and formulation dependent.[10,11,12] There are many analogs available which have higher binding to CB1 receptors with increased psychotropic effects. These are usually

Accepted Article

structurally similar to THC and some have been associated with considerable toxicity. [13]

Interactions between cannabinoids and exercise

Aerobic exercise has a number of effects on the endocannabinoid system. Aerobic exercise results in small elevations in endocannabinoid levels hence a perception that it may be a cause of the "runners high".[14] In regular cannabis users submaximal exercise for 45 minutes results in small elevations of plasma concentrations of THC. These elevations are too small to have a pharmacological effect and are too small to result in misinterpretation of cannabis results detected in drug testing protocols. [15]

Exercise-induced bronchospasm is inhibited by inhaled THC but not CBD. This is of no therapeutic value in the treatment of asthma due to difficulties in formulation, possible adverse psychological effects and the pharmacodynamic superiority of ß2 agonists as bronchodilator agents. [16,17] Aerobic performance and possibly strength is impaired by THC administration. Renaud and Comier found cigarettes with THC 7mg resulted in decreased exercise performance on a bicycle ergometer. [18] Steadward and Singh found that 18.2mg THC impaired submaximal performance using a submaximal bicycle protocol.[19] A further investigation utilising a modified Masters' Step Test and increasing oral doses of THC, found 2 out of 12 healthy males were unable to complete the protocol because of dizziness. [20] Strength has been less well assessed after acute administration but decreased strength of finger flexion has been described by Hollister. [21] Four recently published systematic reviews all considered that administered THC does not improve and probably decreases, aerobic exercise and possibly strength. [22,23,24,25] Excluding sportspeople that have had positive drug tests there is only one report of the effect of cannabis in the situation of high-level competition sport this being one small study finding that cannabis impaired performance in mixed martial arts (MMA). [26]

Chronic cannabis use does not appear to have the same detrimental effect on exercise performance of cannabis users when they are not consuming cannabis at the time. This was first evaluated in 1980 using workers in an industrial setting and quantitated by exercise performance using a bicycle protocol to exhaustion. The participants were either tobacco smokers, marihuana smokers and non-users of either . There was no difference in performance between the three groups. [27] A more recent investigation compared 12 regular users and 12 non-using males aged between 19 and 39 recruited from the Colorado University campus. Cardiorespiratory fitness was measured by Bruce Protocol with measurement of VO_2 max,

Accepted Article

strength by handgrip, anaerobic power by Wingate protocol, and assessments of lower limb strength by Isokinetic Dynamometer. No significant differences were found between the two groups but there was a tendency for greater fatigue in the marihuana users. [28]

Almost all exercise and strength investigations have used young subjects <40 years of age. There are none in children. Of more concern are the findings in two earlier investigations on 10 older patients aged 47.3 +/- 6.1 and 49.6 +/- 5.9 years with angiographically diagnosed >75% narrowing in at least one coronary artery. Smoking cigarettes containing THC 19.8mg prior to exercise resulted in an earlier onset of angina when compared to both placebo and nicotine containing cigarettes which may be due to the earlier increase in peak double product (heart rate x systolic blood pressure).[29,30]

Use in the general community

Cannabis is the most extensively consumed psychoactive drug worldwide. Surveys of the use by sportspeople requires differentiation between use for purely recreational or medicinal consumption and use as an adjunct to sporting activities. With the continuing worldwide liberalisation of laws relating to the use of cannabis presently available figures may be under-estimates of the position in 2022. In the 2016 Australian community cannabis was the most commonly used illicit drug with 10% reporting use in the preceding 12 months. [31]

Use in top level sport

The psychoactive component THC and analogs remain banned substances by the World Anti-Doping Agency (WADA) but CBD is permitted. [32] The urinary concentrations of metabolite THC-COOH detected by WADA accredited laboratories could not be the result of passive exposure and are an indication of use at elite competition level. [33] In a sample covering the period 2014-2017 cannabis constituted around 3% of all positive results the highest use being in weightlifting, wrestling, boxing, cycling, taekwondo, judo and aquatics. [34] In an older survey of 974 elite Australian athletes 3.2% had used cannabis in the past year and 21% had a history of any past use including recreational use. [35] The most recent Australian report of drug testing in the elite did not detect THC in any individual. [36]

Use in lower-level sport

Surveys have been the principal method of determining cannabis use in sportspeople. Sample sizes range from a relatively small number of subjects up to thousands and there is a wide spectrum of results. Published surveys usually comprise of cross-sections of high school and university

Accepted Article

students, sporting organisations, elite athletes, conference attendees and youth surveys. These miss those who are not in organised events or sporting organisations. Almost all respondents are under 40 years old and interpretation of data is complicated by factors such as reluctance to disclose use, details as to the dose and formulation of the active agent(s) used and whether the use relates to some sought after effect in terms of performance or injury. The United States National Health and Nutrition Examination Survey (NHANES) of 20 – 59-year-olds covered the period 2007-2014. After excluding 6,441, 12,618 had complete responses to the "Drug use and exercise" section of the questionnaire (never = 5,913, past = 5,109, current = 1,596). The survey concluded that there was lower prevalence of moderate physical activity in users of marihuana than never users, and there were no differences in the levels of exercise between the samples when adjusted for co-variates. [37] A systematic review of 37 articles on cannabis use by athletes ranging from amateur school level to elite professionals, found usages ranged from 0.8 - 66.8% lifetime use to 2.7 - 62% use in the past year. [24] The studies differed in terms of sport examined, questionnaires used, country and performance level of the participants. One older survey of 1,196 French Sport Science University students found a higher use in those involved in "sliding sports" (wind surfing, skiing, snowboarding, surfing, sailing). [38]

Social media surveys are now an easy means of accessing target groups of interest**.** Since cannabis laws have become relaxed some of these results and opinions have been published in peer reviewed literature. The Athlete-Pain Exercise and Cannabis Experience (PEACE) used Facebook and other social media and flyers in sports stores to access triathletes, swimmers, ultra-runners and cyclists. Respondents were largely serious athletes with 73% exercising >5 days per week and almost half exercising for >11 hours per week. Of the 1161 who completed the survey 67% had used cannabis and 301 (26%) were current users of which 60% were >40 years old with greater use of CBD in the older athlete. Over half used a cannabis preparation (type unspecified) for sleep, calming down and reducing pain. Other responses cited positive and negative effects on performance, decreasing muscle spasms and anxiety/paranoia and increased energy. [39] Another widely advertised online Facebook survey entitled "Cannabis and Health" targeted individuals aged over 21 in states in the USA where cannabis laws had been relaxed. Of the 620 respondents 605 responding to the cannabis questions, with a mean age of 37.5 years old, 81.7% endorsed concurrent use of cannabis (type unspecified) before or after exercise and undertook more exercise than nonusers. The majority of users variously reported it increased enjoyment and improved recovery and motivation to exercise. 37.5% considered it enhanced performance while 62.5%

Accepted Article

disagreed or were neutral on this question. [40] An online survey using two Facebook ultrarunner sites resulted in 609 completed surveys. Of these 8.4% of respondents used a performance enhancing agent during training or competition of which cannabis was used in 48%. [41] Ultrarunning is a relatively new sport defined by competing in events of over 50 km and up to >100 miles.

In summary, at present, cannabis use by athletes appears to be common, but far from universal. Preparations used usually contain both CBD and THC in various proportions. CBD is considered as an adjunct to training, to improve sleep patterns, prevent neurotrauma or relieve musculoskeletal and other discomforts and its use as a single agent is probably higher in the older age groups. THC is considered , among other reported subjective properties, as being able to improve training by increasing the ability to concentrate on form and as having calming effects and improving appetite. Total cannabis use is probably similar to use in the general community.

Adverse effects

There is a very extensive literature relating to adverse effects of cannabinoids but there has not been a comprehensive investigation of adverse effects in the area of sport. Data has been gathered only from surveys and occasional case reports. Individuals who have ceased cannabis because of an adverse experience such as a drug related adverse reaction or an accident caused by incoordination are therefore unlikely to be included in most surveys.

Excluding illegal sources of street cannabis, at present sportspeople in Australia can administer cannabinoids from pharmaceutically formulated combinations either locally available or imported. Formulations contain THC, CBD and occasionally other botanical cannabinoids in various ratios or as separate agents. Many use other preparations where the concentrations are not known and contaminants are sometimes likely to be present and co-administered supplements may also be altering the pharmacokinetics and pharmacodynamics of the cannabinoids. [42]

In volunteer studies single doses of THC produce alterations in mood and some of these such as relaxation and disinhibition appear to be considered a positive adjunct to exercise. In one formal exercise investigation one subject became “stoned” and could not continue the exercise protocol after smoking a single cigarette containing THC 20mg. [43]

Occasionally THC can produce severe paranoia panic attacks and hallucinations. The PEACE investigators reported decreased anxiety,

Accepted Article

calming down effects but also anxiety, paranoia and feelings of uneasiness. [39]

Dose ranging studies of 5, 10, 15 and 20 mg THC administered as capsules cause a dose related decrease in standing steadiness, simple reaction times and visual determination tests lasting up to three hours. [44] These may result in injuries in sports that require high levels of concentration, special awareness and skilled coordination.

Cannabinoids have multiple effects on the cardiovascular system. Single inhaled doses of THC cause an increase in heart rate and blood pressure but in a laboratory investigation where doses of THC were increased in a stepwise fashion, 2 of 12 could not complete the protocol because of dizziness related to falls in blood pressure. [20]

There are a number of case reports of acute cerebrovascular and cardiac events where cannabis is implicated as the causative agent. [45] In the case of patients with coronary disease adverse cardiac effects may occur as the acute elevations of heart rate decrease the threshold for exercise precipitated angina. [29,30] Acute coronary syndromes are rare in the young and in sport but have been attributed to cannabis during sporting activity. [46,47]

Cannabidiol

There is considerable interest in and promotion of this non-psychotropic agent as it now allowed in competition sport. Recent narrative reviews on its potential applications in sport have considered possible positive effects in numerous areas such as inflammation, pain, neuroprotection in contact sport, sleep and anxiety and mentioned the lack of and need for, well-designed studies in sport. [48,49] Such studies can be particularly difficult in sportspersons as a recent meta-analysis found athletes have greater pain tolerance than the sedentary. [50]

There are numerous publications reporting the effects of THC on exercise performance. In contrast there are only three (very recent) publications in the refereed literature that evaluate the effect of CBD administration following either principally upper or lower body exercise.

In the first 23 male and female trained weight lifters were divided into 3 groups and not crossed over. The effect of CBD on lower body exercise was evaluated by delayed muscle soreness (DOMS) induced by a maximal back squat. DOMS was quantitated by a visual analog scale for up to 96 hours post exercise. Immediately post exercise, 8 subjects were

Accepted Article

administered 1 ml of CBD oil 16.67 mg sublingually, 8 subjects were administered 1 ml medium chain triglyceride (MCT) sublingually (SL) and 7 subjects (the Null group) did not have any intervention. DOMS scores increased after exercise in all three groups. The CBD group lifted the largest weights (CBD 102.84 ± 45.15 kg, MCT 82.39 ± 34.44 kg, Null 65.91 ± 16.23 kg) and had the highest DOMS scores post exercise induced muscle damage (CBD 60.3 ± 24.37, MCT 37.99 ± 26.94, Null 25.5 ± 18.34), which fell more rapidly in the CBD group than the MCT and Null groups. The authors concluded there was a greater recovery with CBD but this occurred in the group who lifted considerably heavier weights. [51]

The second was a double-blind crossover study that required multiple attendances and laboratory investigations and was completed by 16 (14 male 2 female) of 21 trained lifters. Strength and power were measured by a maximal back squat and a counter movement maximal jump. Muscle damage was measured by creatine kinase (CK) and myoglobin (Myo) levels at 0, 24, 48 and 72 hours. Immediately post exercise a placebo or CBD 60 mg solubilistat was administered, both administered with 250 ml of water. CBD resulted in decreases in the levels of CK and Myo at the $p<0.05$ level when compared to placebo. There were individual increases and decreases in back squat performance in both groups and no alteration in the countermovement jump. The authors concluded that more data are required for clearer statements in the area. [52]

The third was a placebo-controlled double-blind crossover study completed by 13 untrained males. An isokinetic dynamometer was used to evaluate muscle power. Muscle damage was measured by use of a visual 10-point scale after a series of maximal elbow flexions. Subjects received two doses of 150mg (capsule formulation) or identical placebo containing vegetable oil orally. The first dose was within one hour after exercise and the second 8 hours later. Peak torque measures and the same doses were repeated (a total dose of 300mg daily) at 24, 48 and 72 hours. No differences were found between CBD and placebo in either perceived muscle soreness or power at any time interval. [53]

No adverse effects due to CBD were reported in the three studies and while the methodologies varied, their combined findings suggest CBD has no substantial positive or negative effect on performance, symptomatic muscle damage or damage parameters.

It is very likely CBD would not enhance aerobic performance or strength. Oral doses of 600mg cause slight falls in blood pressure and increases in heart rate in response to hand grip. [54] CBD appears to be free of serious

Accepted Article

adverse effects on a normal cardiovascular system. [55] It does not have the bronchodilator properties of THC. [17]

The survey of 25 competitive rugby union and league teams in the United Kingdom is probably the largest sport related case series involving a single cannabinoid. Of the 517 players 91% were aware of CBD and 133 had or still used it in various doses and formulations. The commonest reasons for use were recovery from pain (80%) and sleep (78%). Lesser reasons were anxiety (32%) and a small number of "medical reasons" which included recovery from concussion. [4]

A comprehensive adverse reaction profile for CBD in animals and humans has recently been published. CBD was administered to both children and adults in varying doses of up to 1000mg per day for different periods of time including around 7 weeks in one study of 88 adult patients with treatment resistant psychoses. The incidence of adverse effects reported ranged from 10-93% with a higher incidence being reported in children with refractory epilepsy. The commonest adverse effects were somnolence, sedation, lethargy and gastrointestinal symptoms such as diarrhoea. Liver function tests may become elevated but it must be remembered that many of the patients were receiving other medications and there is considerable potential for drug interactions with cannabidiol. [56]

Conclusions

In Australia at present cannabis use is probably about the same in sport as it is in the general community though one would expect this to change as promotional activities target this specific market. Formulations of THC and CBD are used in combination or individually with many modes of cannabis delivery. THC and CBD do not appear to be thought of as enhancing performance and surveys of elite sport show their use is well below that of anabolic steroids and stimulants. In relation to adverse effects, anxiety, paranoia palpitations, nausea, vomiting, diarrhoea, wheezing, coughing and decreased performance have been reported on occasions. Apart from sporadic case reports there are at present few data attributing serious adverse effects to cannabis during sporting activity. Acute cardiovascular events appear rare but have been described with THC.

With legalised use in the USA, cannabis is now promoted online and in the general media as an adjunct to sporting activity. In particular, it is promoted to induce calmness and relaxation and decease anxiety and probably to decrease inhibition for more dangerous activities before commencing exercise; to improve focus and concentration during exercise, and to increase feelings of relaxation and relieve musculoskeletal pains and

Accepted Article

muscle stiffness after exercise. None of these parameters, including neuroprotective properties, have been objectively studied in the situation of sport. Older sportspeople and those with known cardiac conditions should be aware of the potential for the development of arrhythmias and acute events. THC and analogs remain banned in elite sport. CBD is allowed but there are virtually no controlled studies of its use in sportspeople. It should be noted that some formulations of CBD may contain sufficient THC to result in a positive test in competition sport. [57] In view of the current milieu of cannabinoids, particularly cannabidiol, there is an urgent need for more high quality preclinical and clinical data to improve our understanding in this area.

REFERENCES

1. Arnold JC, Nation T, McGregor IS. Prescribing Medicinal Cannabis. *Aust Prescr* 2020;43:152-159.

2 . Australian Government Therapeutic Goods Administration. Notice of final decision to amend (or not amend) the current Poisons Standard - cannabidiol. https://www.tga.gov.au/scheduling-decision-final/notice-final-decision-amend-or-not-amend-current-poisons-standard-cannabidiol Accessed 30/8/ 2021

3. Bartels EM, Swadding J, Harrison AP. An ancient Greek pain remedy. *Pain Pract.* 2006;6:212-2118.

4. Kasper AM, Sparks SA, Hooks M, Skeer M et al High prevalence of cannabidiol use within male professional rugby union and league players: A quest for pain relief and enhanced recovery. *Intl J Sport Nut Exerc Metab* 2020; 30: 315-322.

5. Thomas JO, Dunn M, Swift W, Burns l. Illicit drug knowledge and information-seeking behaviour among elite athletes. *J. Science and Med in Sport*. 2011; 14 (4): 278-282.

6. Haddad S. n.d., Could getting stoned actually make you bertter at sports? This Olympic gold medallist thinks so… *mpora* https://mpora.com/snowboarding/cannabis-performance-enhancing-drug-Ross-Regabkiati-interview accessed 16/1/2022

7. Hasse J. USA Triathlon becomes First U.S. Olympic and Paralympic committee member to Ink CBD Partnership. *Forbes Magazine* Oct 31,

Accepted Article

2019 https://www.forbes.com/sites/javierhasse/2019/10/31/usa-triathlon-cbd/?sh=1ef878932b4c accessed 22/1/2022

8. Cannabidiol brand will sponsor World Surf League. *Surfer Today.* Nov 5 2019: 278-282

9. Andre C, Hausman J-F, Guerriero G. Cannabis sativa: The Plant of the Thousand and One Molecules. *Front. Plant Sci.* 2016: 7:19

10. Grotenhermen F. Pharmacokinetics and pharmacodynamics of cannabinoids. *Clin Pharmacokinet,* 2003;42(4): 328-360

11. Zendulka Q, Dovrtelova G, Novskova K, Turjap M, et al. Cannabinoids and cytochromeP450 interactions *Curr Drug Metab.* 2016;17(3): 206-226.

12. Millar SA, Stone N, Tates AS, O'Sullivan S. A systematic review of cannabidiol in humans. *Front Pharmacol,* 2018;9:1-13

13. Ford BM, Tai S, Fantegrossi WE, Prathher PL. Synthetic pot: Not your grandfather's marijuana. *Trends Pharmacol Sci.* 2017;38(3): 257-276.

14. Heyman E, Gamelin F-X, Goekint M, Piscitelli F et al Intense exercise increases circulating endocannabinoid and BDNF levels in humans - Possible implications for reward and depression *Psychoneuroendocrinology* 2012;37:844-851

15. Wong A, Montebello ME, Norberg MN , Rooney K, et al Exercise increases plasma THC concentrations in regular cannabis users. *Drug alcohol depend* 2013;133;763-767.

16. Tashkin DP, Bertrand J , Lee E, Harper CE. Effects of smoked marijuana in experimentally induced asthma. *Am Rev Respir Dis* 1975;112:377-386.

17. Gong H, Tashkin DP, Simmons BA, Calvarese B,etal . Acute and subacute bronchial effects of oral cannabinoids *Clin Pharmacol Ther* 1984;34(1):27-32.

18. Renaud AM, Cormier Y. Acute effects of marijuana smoking on maximal exercise performance. *Med Sci Sports Exerc* 1986;18(6):685-689.

Accepted Article

Accepted Article

19. Steadward RD, Singh M. The effects of smoking marijuana on physical performance. *Med Sci Sports Exerc* 1975;7(4):309-311.

20. Benowitz NL, Jones R. Cardiovascular effects of prolonged delta-9-tetrahydrocannabinol ingestion *Clin Pharmacol Ther* 1975;18(3):287-297.

21. Hollister LE, Richards KR, Gillespie BA. Comparison of tetrahydrocannabinol and synhexal in man. *Clin Pharmacol Ther* 1968;5(76); 783-791.

22. Kennedy MC. Cannabis: Exercise performance and sport. A systematic review. *J Sci Med Sport* 2917;20:825-829.

23.Trinh KV, Diep D, Robson H. Marijuana and its effects on athletic performance: A systematic review. *Clin Sports Med* 2018;28(4): 350-

24. Docter SBK. In Kahn M, Gohal C, Ravi B, Bhandari M, Gandhi R, Leroux T. Cannabis use and sport: A systematic review. *Sports Health 2020*;12(2): 189-199.

25. Kramer A, Sinclair J, Sharpe LK, Sarris. Chronic cannabis consumption and physical exercise performance in healthy adults : A systematic review *J Cannabis Res* 2020;2:34

26. Aedo-Munoz E, Broto CJ, Antonietto NR, Duarte MB, Mozer RL, Coswig VS. Adverse effects of cannabinoids on behaviours of martial arts professional athletes :an ex-postfacto study. *Motricidade* 2019;15(nS3): 28-32.

27. Maksud MG, Baron A. Physiological responses to exercise in chronic cigarette and marijuana users. *Eur J Appl Physiol* 1980;43:127-134.

28. Lisano JK, Smith JD, Alissa AB, Christensen M, Somak P, Phillips T, Quinn CJ, Stewart LK. Performance and health -related characteristics of physically active males using marihuana. *J Strength and Cond Res* 2019;33(6): 1658-1668

29. Aronow WS, Cassidy J. Effect of marihuana and placebo-marihuana smoking on angina pectoris. *N Engl J Med* 1974;291(2):675-67. doi:

30. Aranow WS, Cassidy J. Effect of smoking marijuana and a high-nicotine cigarette on angina pectoris. *Clin Pharmacol Ther 1974*;17(5):549-554.

31. Australian Government Institute of Health and Welfare. National drug strategy household survey 2016. https://www.aihw.gov.au/reports/illicit-use-of-drugs/2016-ndshs-detailed/summary accessed 2/9/21

32. World Anti-Doping Code International Standard Prohibited list 2021 https://www.wada-ama.org/en/what-we-do/the-prohibited-list . accessed 2/9/21

33. Huestis MA. Human Cannabinoid pharmacokinetics. *Chem Biodivers* 2007;4(8): 1770-1804.

34. Aguilar-Navarro M, Salinero JJ, Munoz-Guerra J, Maria Del Mar Plata M, et al J Sport-specific use of doping substances :Analysis of world-anti-doping agency doping control tests between 2014 and 2017 *Subst Use Misuse* 2020;55(8): 1361-1369

35.Dunn M, Thomas JO, Swift W, Burns L. Recreational substance use among elite Australian athletes. *Drug Alcohol Rev* 2011;30(1);63-68.

36. Sport Integrity Australia. Annual Report 2019-2020. https://www.sportintegrity.gov.au/sites/default/files/10784%20ASADA%20Annual%20Report%202019%C2%AD20_FA-tagged-for-web.pdf accessed 2/9/21

37. Vidot DC, Bispo JD, Hlaing WM, Prado G et al Moderate and vigorous physical activity patterns among marijuana users: Results from the 2007-2014 National Health and Nutrition examination surveys. *Drug Alcohol Depend* 2017;178: 43-48.

38. Lorente FO, Pretti-Watel P, Grelot L. Cannabis use to enhance sportive and non-sportive performances among French sport students. *Addict Behav* 2005. 30:1382-1391.

39. Zeigler JS, Sikvers WS, Fleeger EM, Zeigler RS Cannabis use in active subjects: Behaviours related to subjective effects *PLoS ONE* 14(6):e0218998

40. YorkWilliams S, Gust CJ, Mueller R, Bidwell LC, et al A new runners high? Examining relationships between Cannabis use and

Accepted Article

exercise behaviour in states with legalised cannabis *Public Health Front* 2019;7:article 99.

41. Campian MD, Alexandra EF, Teramoto M, Cushman DM. Self-reported use and attitudes toward performance-enhancing drugs in ultramarathon running. *Wilderness Environ Med* 2018;29: 330-337.

42. Dryburgh LM, Bolan NS, Grof CPL, Galettis P, Schneider J, Lucas CJ, Martin J. Cannabis contaminants: sources, distribution, human toxicity and pharmacological effects. *Br J Clin Pharmacol* 2018. 84: 2468-2476

43. UCLA Conference: Cannabis 1977. Cardiovascular Effects of Marijuana. Dr Bertrand Shapiro. *Ann Intern Med* 1978;89: 544-546

44. Chesher GB, Bird KD, Jackson DM, Perrignon A et al The effects of orally administered Δ9-tetrahydrocannabinol in man on mood and performance measures: A does-response study. *Pharmacol Biochem Behav* 1990;13: 861-864.

45. Goyal H, Awad H, Ghali JK. Role of cannabis in cardiovascular disorders. *J Thorac Dis* 2017;9(7) 2079-2092.

46. Deharo P, Massoure P-L, Fourcade L. Exercise-induced acute coronary syndrome in a 24-year-old man with massive cannabis consumption *Acta Cardiol* 2013;68(4): 425-428

47. Hodcroft CJ, Rossiter MC, Buch AN. Cannabis-associated myocardial infarction in a young man with normal coronary arteries. *J Emerg Med* 2014;47(3):277-281.

48. McCartney D, Benson MJ, Desbrow B, Irwin C Suraev A,Mc Gregor IS
Cannabidiol and sports performance: A narrative review of relevant evidence and recommendations for future research *Sports Medicine-open* 2020;6:27

49. Gamelin F-X, Cuvelier G, Mendes A, Aucouturier J,Berthoin,DiMarzo V,Heyman E Cannabidiol in sport :Ergogenic or else? *Pharmacol Res* 2020;156:104764

Accepted Article

50. Tesarz J, Schuster AK, Hartmann M, Gerhart A, Eich W. Pain perception in athletes compared to normally active controls: A systematic review and meta-analysis. *Pain* 2012; 153: 1253-12625.

51. Hatchett A, Armstrong K, Hughes B, Parr B. The influence of cannabidiol on delayed onset of muscle soreness. *Inter J Phys Educ, Sports Health* 2020;7(2): 89-94.

52. Isenmann E, Veit S, Starke L, Flenker U e tal Diel F. Effects of cannabidiol supplementation on skeletal muscle regeneration after intensive resistance training. *Nutrients* 2021;13:3028.

53. Cochrane-Synman KC, Cruz C, Morales J, Coles M. The effects of cannabidiol oil on noninvasive measures of muscle damage in men. *Med Sci Sports Exerc* 2021; 53(7):1460-1472.

54. Jadoon KA, Tan GD, O'Sullivan SE. A single dose of cannabidiol reduces blood pressure in healthy volunteers in a randomised crossover study. *JCI Insight* 2017;2:12.

55. Sultan SR, Millar SA, England J, O'Sullivan EO. A systematic review and meta-analysis of haemodynamic effects of cannabidiol. *Front Pharmacol* 2017;8:81.

56. Huestis MA, Solimini R, Pichini S, Pacifici R et al Cannabidiol adverse effects and toxicology. *Curr Neuropharmacol* 2019;17:974-978.

57. Lachenmeier DW, Diel P. A warning against negligent use of cannabidiol in professional and amateur athletes. *Sports* 2019;7(12)

Accepted Article